

MOHAMMAD RAIHANUL BASHAR

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Research Interest

I am a PhD candidate at Concordia University. My research lies at the intersection of **Human-Computer Interaction**, Computer Graphics, and Computer Vision. My goal is to **empower human imagination and cognition** through XR and AI by enabling the world itself to function as an interactive medium for exploration and expression.

Education

- 01/2024 – **Ph.D. in Computer Science (ABD)**
current Department of Computer Science & Software Engineering, Concordia University
Supervisor: Dr. Anil Ufuk Batmaz
- 09/2023 – **MSc in Computer Science**
12/2023 Department of Computer Science & Software Engineering, Concordia University
Fast-track to PhD
Supervisor: Dr. Anil Ufuk Batmaz
- 01/2013 – **BSc in Computer Science**
08/2017 Department of Computer Science, Independent University, Bangladesh
Effect of EMG Artifacts on Video Category Classification from EEG
Advisor: Prof. M. Ashraful Amin

Publications

Papers with future publication dates are accepted to appear in their respective venues.

Book Chapters

- 2017 [B. 1] Computational Intelligence for Pattern Recognition in EEG Signals. Aunnoy K Mutasim, Rayhan Sardar Tipu, **Mohammad Raihanul Bashar**, Md Kafiul Islam, M. Ashraful Amin. *Computational Intelligence for Pattern Recognition*, Springer-Verlag.

Peer-Reviewed Hybrid Conference-Journal Articles

- 2025 [J. 4] An Early Warning System Based on Visual Feedback for Light-Based Hand Tracking Failures in VR Head-Mounted Displays. **Mohammad Raihanul Bashar**, Anil Ufuk Batmaz. (*TVCG (IEEE VR' 25)*).

Peer-Reviewed Journal Articles

- 2026 [U-] Play and Learn: VR Catheter Insertion Training. Bashar MR et al. (in submission to JMIR SG)
- 2025 [J. 3] The Effect of Visual Depth on the Vergence-Accommodation Conflict on 3D Selection Performance within Virtual Reality Headsets. **Mohammad Raihanul Bashar**, Mayra Donaji Barrera Machuca, Wolfgang Stuerzlinger, Anil Ufuk Batmaz. *The Visual Computer, Springer-Verlag*.
- 2023 [J. 2] EEG-Based Preference Classification for Neuromarketing Application. Injamamul Haque Surov, Faiyaz Alvi Ahmed, Md. Tawhid Islam Opu, Aunnoy K Mutasim, **M. Raihanul Bashar**, Rayhan Sardar Tipu, M Ashraful Amin, and Md Kafiul Islam. *Computational Intelligence and Neuroscience*.
- 2020 [J. 1] Use of spontaneous blinking for application in human authentication. Amir Jalilifard, Dehua Chen, Aunnoy K Mutasim, **M. Raihanul Bashar**, Rayhan Sardar Tipu, Md. Ahsan-Ul Kabir Shawon, Nazmus Sakib, M Ashraful Amin, and Md Kafiul Islam. *Engineering Science and Technology, an International Journal, Elsevier*.

Peer-Reviewed Conference Proceedings

- 2026 [U-] Multimodal Techniques for Multi-Object Selection. Bashar MR et al. (in submission to CHI)
- 2026 [U-] Multimodal Situated Interaction. Bashar MR et al. (in submission to CHI)
- 2025 [C.7] There Is More to Dwell Than Meets the Eye: Toward Better Gaze-Based Text Entry Systems With Multi-Threshold Dwell. Aunnoy K Mutasim, **Mohammad Raihanul Bashar**, Christof Lutteroth, Anil Ufuk Batmaz, Wolfgang Stuerzlinger. (*CHI' 25*).
- 2024 [C.6] Virtual Task Environments Factors Explored in 3D Selection Studies. **Mohammad Raihanul Bashar**, Anil Ufuk Batmaz. (*ACM GI' 24*).
- 2018 [C. 5] Effect of Artefact Removal Techniques on EEG Signals for Video Category Classification. Aunnoy K Mutasim, **Mohammad Raihanul Bashar**, Rayhan Sardar Tipu, Md Kafiul Islam, M. Ashraful Amin. 24th *International Conference on Pattern Recognition (ICPR' 18)*.
- 2018 [C.4] Effect of EMG Artifacts on Video Category Classification from EEG. **Mohammad Raihanul Bashar**, Rayhan Sardar Tipu, Aunnoy K Mutasim, M. Ashraful Amin. (*ICIEV*) & (*icIVPR*).

- 2017 [C.3] Video Category Classification Using Wireless EEG. Aunnoy K Mutasim, Rayhan Sardar Tipu, **Mohammad Raihanul Bashar**, M. Ashraful Amin. (*BI' 17*).
- 2017 [C.2] Implementation of Low Cost Stereo Humanoid Adaptive Vision for 3D positioning and Distance Measurement for Robotics Application with Self-calibration. Abul Al Arabi, Rayhan Sardar Tipu, **Mohammad Raihanul Bashar**, Binoy Barman, Shama Ali Monica, M. Ashraful Amin. (*AMS' 17*).
- 2017 [C.1] 2D Surface Mapping for Mine Detection via Wireless Networking. **Mohammad Raihanul Bashar**, Abul Al Arabi, Rayhan Sardar Tipu, Md Tanvir Alam Sifat, Md Zobair Ibnalam, M. Ashraful Amin. (*ICCRE' 17*).

Peer-Reviewed Conference Poster Abstracts

- 2025 [A.1] Depth3DSketch: Freehand Sketching Out of Arm's Reach in Virtual Reality. **Mohammad Raihanul Bashar**, Mohammadreza Amini, Wolfgang Stuerzlinger, Mine Sarac, Ken Pfeuffer, Mayra Donaji Barrera Machuca, Anil Ufuk Batmaz. (*CHI EA' 25*).

Peer-Reviewed Workshop Proceedings

- 2024 [W.1] Subtask-Based Virtual Hand Visualization Method for Enhanced User Accuracy in Virtual Reality Environments. Laurent Voisard, Amal Hatira, **Mohammad Raihanul Bashar**, Mucahit Gemici, Mine Sarac, Marta Kersten-Oertel, Anil Ufuk Batmaz. (*NIDIT' 24*).

Research Experience

- 12/2022 – *current* **EXiT Lab**, Concordia University, Graduate Student Researcher. Montreal, QC. Advisor: Dr. Anil Ufuk Batmaz. Driving research agenda on pushing the frontier of human interaction in eXtended Reality by developing interaction systems that empower human creativity. Publishing at top-tier HCI conferences.
- 01/2019 – 12/2021 **AGenCy Lab, IUB**, Independent University, Bangladesh, Research Assistant. Dhaka, BD. Advisor: Prof. M. Ashraful Amin. Develop a high-quality, standardized medicine image dataset with metadata, deploy a recognition model to enhance feature selection, and create a mobile application to assist visually impaired patients in identifying their prescribed pills. Work led to opensource code [O. 3].
- 01/2018 – 12/2019

Computer Vision & Cybernetics, BD, Independent University, Bangladesh, Research Assistant. Dhaka, BD. Advisor: Prof. M. Ashraful Amin. The goal was to compile a dataset of Bengali letters and numbers for Bangladeshi license plates and develop a custom ANPR system, successfully addressing the lack of open-source datasets and enabling accurate recognition of Bangladeshi license plates. Work led to opensource code [O. 2].

04/2016 – 01/2018 **Computer Vision & Cybernetics, BD**, Independent University, Bangladesh, Undergraduate Research Assistant. Dhaka, BD. Advisor: Prof. M. Ashraful Amin. With the vision of building a BCI-based video recommender system, we experimented with several state-of-the-art algorithms for each of the submodules (pre-processing, feature extraction, feature selection, and classification) of the Signal Processing module of a BCI system to find out what combination of algorithms best predicts what type of video a person is watching. Work led to publication at BI 2017 [C. 3], ICIEV 2017 [C. 4], ICPR 2018 [C. 5] and a Book Chapter [B. 1] and opensource code [O. 1].

Teaching

Courses

Winter 2026 **SOEN 287: Web Programmimg (Undergraduate)** Department of Computer Science and Software Engineering, Concordia University

Teaching Assistant

Fall 2025 **COMP 5201: Computer Organization and Assembly Language(Graduate)**
Instructor: Dr. David K. Probst
Department of Computer Science and Software Engineering, Concordia University

Fall 2025 **COMP 472: Introduction to Artificial Intelligence (Undergraduate)**
Instructor: Dr. Kefaya Qaddoum
Department of Computer Science and Software Engineering, Concordia University

Fall 2023, 2024, 2025 **COMP 376: Introduction to Game Developments (Undergraduate)**
Instructor: Dr. Anil Ufuk Batmaz
Department of Computer Science and Software Engineering, Concordia University

Winter 2024 **COMP 6331: Advanced Game Developments (Graduate)**
Instructor: Dr. Anil Ufuk Batmaz
Department of Computer Science and Software Engineering, Concordia University

Winter 2024 **SOEN 6751: Human-Computer Interaction (Graduate)**
Instructor: Dr. Anil Ufuk Batmaz
Department of Computer Science and Software Engineering, Concordia University

- Fall 2018 **CSE 420: Image Processing & Pattern Recognition, CSE 420. (Undergraduate)**
Instructor: Prof. M. Ashraful Amin
Department of Computer Science and Engineering, Independent University, Bangladesh.
- Fall 2018 **CSE 417: Data Mining and Warehouse (Graduate)**
Instructor: Dr. Moinul Zaber
Department of Computer Science and Engineering, Independent University, Bangladesh.

Industry Experience

- Fall 2024 **Mitacs Accelerate Internship**, Research Intern. Remote. SonoNurse Inc., Concordia University, Montreal, QC.
- 05/2022 – **Senior Engineer, MLOps.** Quantigo AI, Dhaka, Bangladesh. Led the development and launch of two web applications, including a workforce management system, while providing customized ML-assisted data annotation solutions, developing an in-house video annotation platform, and establishing foundational architectures for current and future projects.
- 08/2023
- 07/2021 – **Senior AI Engineer.** OPUS Technology Limited, Dhaka, Bangladesh. Developed an intelligent agent for CS:GO using offline RL, supervised the creation of an e-sports game prediction system, implemented a matchmaking system with AWS Gamelift, streamlined MLOps integration with cloud platforms, mentored the ML team on RL and clean coding practices, and maintained the company's largest project while contributing reusable modular-helper classes.
- 03/2022
- Summer 2021 **Machine Learning Engineer.** Expert Consortium Limited, Dhaka, Bangladesh. Developed an early shoplifting detection system.
- 11/2020 – **Software Engineer.** Maxis Systems Limited, Dhaka, Bangladesh. Integrated a DevOps pipeline as the development platform, developed scalable and maintainable microservice solutions, and built a platform for business analytics leveraging ML and Data Science.
- 03/2021

Awards and Scholarships

- 2025 **Concordia Conference and Exposition Allowance, CAD 1000**, School of Graduate Studies, Concordia University
- 2025 **Concordia Conference and Exposition Allowance, CAD 1000**, School of Graduate Studies, Concordia University

2024	Concordia Conference and Exposition Allowance, CAD 1000, School of Graduate Studies, Concordia University
2024	FRS Scholarship for PhD Student, CAD 22,000/year, Concordia University
2024	Concordia International Tuition Award of Excellence, CAD 44,893, Concordia University
2023	FRS Scholarship for MSc Student, CAD 24,000/year, Concordia University
2023	Concordia Merit Scholarship, CAD 5,000, Concordia University
2017	SECS Award, Independent University, Bangladesh

Service

Service items with future dates are confirmed commitments.

ORGANIZING

2026	Program Committee Member. ACM IUI 2026.
2025	Program Committee Member. ACM MUM 2025.
2025	Student Volunteer Chair. ACM SUI 2025.
2025	Student Volunteer. IEEE VR 2025.

REVIEWING

2026	Reviewer. The ACM Conference on Human Factors in Computing Systems (CHI)
2025	Reviewer. Virtual Reality Software and Technology (VRST)
2025	Reviewer. The ACM Symposium on User Interface Software and Technology (UIST)
2025	Reviewer. IEEE International Symposium on Mixed and Augmented Reality (ISMAR)
2025	Reviewer. The ACM Conference on Human Factors in Computing Systems Late-Breaking Work (CHI LBW)
2025	Reviewer. The IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)
2024	Reviewer. The ACM Conference on Human Factors in Computing Systems (CHI)
2024	Reviewer. The ACM Conference on Mobile Human-Computer Interaction (MobileHCI)
2024	Reviewer. The IEEE International Symposium on Mixed and Augmented Reality (ISMAR)
2024	Reviewer. The ACM Symposium on Spatial User Interaction (SUI)

COMMUNITY

2025	Presenter. Girls Summer Engineering and Technology Program (GirlSET), Concordia University.
2024	Presenter. Girls Summer Engineering and Technology Program (GirlSET), Concordia University.

Skills

I have expertise in developing interactive XR and AI systems, as well as full-stack web development, encompassing both frontend and backend. Additionally, I have experience in designing, training, and evaluating deep learning models. I am familiar with the following programming languages, frameworks, and research methodologies.

Programming Languages: Python, C++, C#, HTML/CSS/JavaScript

Frameworks: PyTorch, Scikit-Learn, Unity, Unreal, OpenXR, Flask, React

Research Methodologies: User Studies (quantitative/qualitative), DL experiments.

References

Available upon requests.